PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effect of Home-based Interventions on Basic Activities of Daily Living for Patients with Stroke: A Systematic Review with Meta-analysis
AUTHORS	Qin, Ping; Cai, Canxing; Chen, Xuan; Wei, Xijun

VERSION 1 – REVIEW

REVIEWER	Sheehy, Lisa Mary Bruyère Research Institute
REVIEW RETURNED	04-Oct-2021

GENERAL COMMENTS	In this systematic review and meta-analysis, the authors studied the effect of home-based interventions on basic activities of daily living in patients with stroke. They compared home-based interventions with institutional-based interventions, usual care, home-based interventions + usual care, and no intervention. In the meta-analysis, only the comparison of home-based interventions compared to home-based interventions + usual care yielded a small but significant result. General comments: This is a well-done study. The methods and results are appropriate and clearly presented. The tables and figures are useful and clear. The importance of this topic is high, and this will inform clinicians who use home-based stroke rehabilitation about their practice. Probably the greatest use, however, will be for researchers, to direct future research studies. While the quality of the English is good, please proof-read the entire manuscript. Specific Comments: Results and Figures 2-5 The inclusion of weighting related to study quality is appreciated. At first I was wondering where the Summary Measures were on the forest plots, but I see that they are just too small to visualize. I2=0% is found in several places, ex. Pg 44, line 38. Is this correct? You mention that “statistical heterogeneity decreased”, but decreasing from 64% to 0% seems unlikely. This type of comment is repeated in this section. Is it possible to have CIs around I2? Figure 5 – Wong et al. does not look correct – the IV of 0.66 says that the CI is 0.27-1.04 so the green box should not be over the “0” line. But it is.

	Limitations Add a mention about the limitation of only including 3-4 studies in a meta-analysis. Discussion Some additions could be made to the discussion, for example, a discussion on the content of “usual care” and how that might have affected the results in the different studies. For example, if usual care is very close to the intervention, did that produce poorer results than if usual care was minimal? A discussion could be made of the impact of the addition of home-based intervention to usual care, compared to usual care. Since this was the only comparison that yielded statistically positive results, why might this be so? I postulate that in these studies the rehabilitation intensity would have been greater in the intervention group, leading to greater response (i.e. more exercise overall leads to better outcomes). This would not have occurred in the other comparisons. Please include in the discussion. In the conclusions, a specific plea as to what future research should include (ex. Larger sample sizes, particular outcomes) would be a nice addition to this section. The Stroke Recovery and Rehabilitation Round Table has some guidelines on choosing outcomes for stroke research, so that in the future, research results will be able to be compared between studies. https://journals.sagepub.com/page/wso/srrr I noticed that in several instances, there were two manuscripts that might have reported on data from the same study. For example, Kalra 2020 and Kalra 2005; Gladman 1993 and Gladman 1994; Thorsen 2005, Viden 1998 and von Koch 2000; Chen 2017 and Chen 2020; Kulkantrak (2012a and b). I see that in no instance did you include data from two potentially related studies in your meta-analyses, but this might deserve a mention in the discussion. You report 47 manuscripts, but this might actually represent 41 or fewer studies.

REVIEWER	Banno, Masahiro Seichiryo Hospital, Department of Psychiatry
REVIEW RETURNED	21-Dec-2021

GENERAL COMMENTS	This study aimed to evaluate the effectiveness of home-based interventions on the performance of basic activities of daily living in stroke patients when compared with institution-based intervention, usual care, and no intervention respectively. The strength of this study was to conduct analysis using a rigorous method of systematic reviews. However, there was some weakness in this study. First, the authors had better conduct the update search for all the databases (for example, the new search date is December 31, 2021). Second, they had better adhere to PRISMA 2020 for Abstract. For example, they had better describe the following information in the abstract: exclusion criteria for the review, total number of participants, relevant characteristics of included studies, indicating the number of included studies and participants for each primary outcome, the primary source of funding for the review, the register name and registration number.

	Third, it would be better to add the PRISMA 2020 Checklist, not PRISMA 2009 Checklist, as Supplemental Materials with the page numbers of the manuscript for each item in the PRISMA 2020 Checklist. Fourth, they had better revise the manuscript, or add new limitations in the discussion if any important items in the PRISMA 2020 checklist were not met. Fifth, they had better clarify PRISMA 2020 flow diagram in Figure 1. The present Figure 1 was the PRISMA flow diagram in the previous version, not PRISMA 2020 flow diagram. Sixth, they had better cite the article about PRISMA 2020 statement (Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, McKenzie JE. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. 2021 Mar 29;372:n160. doi: 10.1136/bmj.n160.), not the PRISMA statement in 2009. Furthermore, they had better describe they adhered to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 reporting guideline in METHODS.

REVIEWER	Han, Dug Yeo Auckland District Health Board, Starship Child Health
REVIEW RETURNED	19-Jan-2022

GENERAL COMMENTS	It is meta-analysis. No suggestion made for statistical analysis.

REVIEWER	HUGUES, AURELIEN Service de médecine physique et réadaptation, hôpital Henry-Gabrielle, Hospices Civils de Lyon
REVIEW RETURNED	22-Feb-2022

GENERAL COMMENTS	Review The topic of the research is relevant. The method used is adapted. But the meaning given to the results by the authors is questionable. General Avoid “stroke patients” and rather prefer patients with stroke or patients after stroke Abstract I2 is missing for short-term effects of home-based rehabilitation combined with UC versus UC alone. The first sentence of the conclusion is questionable “Home-based rehabilitation may be an alternative to institution-based rehabilitation and usual care for stroke patients.” Methodological concerns noted later in the conclusion and the request of future researches argue against this statement. Introduction The background is well documented and the justification of research is well established.

	Method The search is a bit old (until 31/12/2020). Why? The authors should update the search. The authors do not report which method was applied to limit publication bias? Was the grey literature searched? More details could be provided on data extraction, in particular for the point (2). Please specify the method used to estimate summary effect? Inverse-variance method I think; all variables were continuous outcomes? For the first subgroup analyses, the authors could specify that the level of disability of patients is that in baseline. The authors could justify the cut-off chosen (3 points) to consider poor methodological quality of the studies by including a reference. Results Study characteristics: Maybe the authors could indicate, in brackets, the total number of participants for each category of comparisons. Moreover, the authors could shortly describe the population/participants included in this review (age, time post-stroke, gender, ...) The table 1 is difficult to read due to its length. It would be better to shorten it a bit. The categorization for the disability level of stroke is not always defined in table 1. “Effectiveness of home-based intervention compared with institution-based intervention” part : typo on “a fix-effect” in the last sentence. Figures 2,3,4 and 5: please adjust the scale of the forest plot. Discussion and conclusion Given the results reported, conclude that the effects between home-based intervention and institution-based intervention or usual care might be equivalent seems overestimated. The results such as presented herein do not support these statements. They seem to be conflicting with the limitations such as reported by the authors. “There is no sufficient study to determine the effectiveness of home-based intervention comparing with other approaches”.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Lisa Mary Sheehy, Bruyère Research Institute

Comments to the Author:
In this systematic review and meta-analysis, the authors studied the effect of home-based interventions on basic activities of daily living in patients with stroke. They compared home-based interventions with institutional-based interventions, usual care, home-based interventions + usual care, and no intervention. In the meta-analysis, only the comparison of home-based interventions compared to home-based interventions + usual care yielded a small but significant result.
Reply: Thank you Dr. Lisa Mary Sheehy for your read in detail.

General comments:
This is a well-done study. The methods and results are appropriate and clearly presented. The tables and figures are useful and clear. The importance of this topic is high, and this will inform clinicians who use home-based stroke rehabilitation about their practice. Probably the greatest use, however, will be for researchers, to direct future research studies.

Reply: Thank you Dr. Lisa Mary Sheehy for the comments.

While the quality of the English is good, please proof-read the entire manuscript.
Reply: Spelling and grammar errors throughout the text had been revised accordingly.

Specific Comments:

Results and Figures 2-5
The inclusion of weighting related to study quality is appreciated. At first I was wondering where the Summary Measures were on the forest plots, but I see that they are just too small to visualize.
Reply: The size of forest plots has been adjusted.

I2=0% is found in several places, ex. Pg 44, line 38. Is this correct? You mention that “statistical heterogeneity decreased”, but decreasing from 64% to 0% seems unlikely. This type of comment is repeated in this section. Is it possible to have CIs around I2?
Reply: As far as I know, there is no CIs available around I2. The I2 was automatically calculated by Revman based on the data extracted from the included articles.

Figure 5 – Wong et al. does not look correct – the IV of 0.66 says that the CI is 0.27-1.04 so the green box should not be over the “0” line. But it is.
Reply: Thank Dr. Lisa Mary Sheehy for your read in detail. It had been revised.

Limitations
Add a mention about the limitation of only including 3-4 studies in a meta-analysis.
Reply: The limitation had been mentioned.

Discussion
Some additions could be made to the discussion, for example, a discussion on the content of “usual care” and how that might have affected the results in the different studies. For example, if usual care is very close to the intervention, did that produce poorer results than if usual care was minimal?
Reply: That is a good direction for discussion, but currently it's difficult to make such speculation as the individual studies usually did not specify the detailed information in usual care, such as the dosage, duration...

A discussion could be made of the impact of the addition of home-based intervention to usual care, compared to usual care. Since this was the only comparison that yielded statistically positive results, why might this be so? I postulate that in these studies the rehabilitation intensity would have been greater in the intervention group, leading to greater response (i.e. more exercise overall leads to better outcomes). This would not have occurred in the other comparisons. Please include in the discussion.
Reply: Good suggestion, it had been added in the first paragraph of Discussion.

In the conclusions, a specific plea as to what future research should include (ex. Larger sample sizes, particular outcomes) would be a nice addition to this section. The Stroke Recovery and Rehabilitation Round Table has some guidelines on choosing outcomes for stroke research, so that in the future, research results will be able to be compared between studies.
https://journals.sagepub.com/page/wso/srrr
Reply: Good comments, it had been added in the Conclusion.

I noticed that in several instances, there were two manuscripts that might have reported on data from the same study. For example, Kalra 2020 and Kalra 2005; Gladman 1993 and Gladman 1994; Thorsen 2005, Viden 1998 and von Koch 2000; Chen 2017 and Chen 2020; Kulkantrak (2012a and

b). I see that in no instance did you include data from two potentially related studies in your meta-analyses, but this might deserve a mention in the discussion. You report 47 manuscripts, but this might actually represent 41 or fewer studies.
Reply: Good comments. After read through the articles, we made the revision as following.
Kalra 2005 was a cost-effectiveness study employed the same population as Kalra 2000. Thus, the Kalra 2005 was deleted from the review during the study selection process.
Gladman 1994 was a follow-up study of the trial in Gladman 1994, and the same study population was used, this had been clarified in the limitation in current review. Thorsen 2005, von Koch 2001, von Koch 2000 were the follow-up studies of the same trial in Widen Holmquist 1998. This had been clarified in the limitation in current review.
Chen 2017 and Chen 2020 were two different trials with different study population.
Chaiyawat 2012a and Chaiyawat 2012b were duplicate publication in two different journals. Thus, we deleted Chaiyawat 2012b (Randomized controlled trial of home rehabilitation for patients with ischemic stroke: impact upon disability and elderly depression) during the study selection process. Chaiyawat 2012a (Effectiveness of home rehabilitation program for ischemic stroke upon disability and quality of life: A randomized controlled trial) was a follow-up study of the trial in Chaiyawat 2009, which shared the same study population.
Therefore, there were 44 trials among the 49 were included. This had been clarified in the limitation part in the current review.

Reviewer: 2
Dr. Masahiro Banno, Seichiryo Hospital, Nagoya University Graduate School of Medicine

Comments to the Author:
This study aimed to evaluate the effectiveness of home-based interventions on the performance of basic activities of daily living in stroke patients when compared with institution-based intervention, usual care, and no intervention respectively. The strength of this study was to conduct analysis using a rigorous method of systematic reviews. However, there was some weakness in this study.
Reply: Thank you Dr. Masahiro Banno for your read in detail.

First, the authors had better conduct the update search for all the databases (for example, the new search date is December 31, 2021).
Reply: This study had been updated to December 31, 2021.

Second, they had better adhere to PRISMA 2020 for Abstract. For example, they had better describe the following information in the abstract: exclusion criteria for the review, total number of participants, relevant characteristics of included studies, indicating the number of included studies and participants for each primary outcome, the primary source of funding for the review, the register name and registration number.
Reply: Good comments. We had added the relevant information, except for the register name and registration number as we did not register for the review. The total number of participants was not mentioned in the abstract, as we had conducted several meta-analyses according to the different comparisons. Therefore, the number of participants included in each meta-analysis was not the same and needed detailed clarification which was shown in the main text.

Third, it would be better to add the PRISMA 2020 Checklist, not PRISMA 2009 Checklist, as Supplemental Materials with the page numbers of the manuscript for each item in the PRISMA 2020 Checklist.
Reply: Good comments. We had updated to the PRISMA 2020 Checklist.

Fourth, they had better revise the manuscript, or add new limitations in the discussion if any important items in the PRISMA 2020 checklist were not met.

Reply: The publication bias and the limited number of included studies in each meta-analysis had been added to methodological limitation.

Fifth, they had better clarify PRISMA 2020 flow diagram in Figure 1. The present Figure 1 was the PRISMA flow diagram in the previous version, not PRISMA 2020 flow diagram.
Reply: We had updated to the PRISMA 2020 flow diagram in Figure 1.

Sixth, they had better cite the article about PRISMA 2020 statement (Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, McKenzie JE. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. 2021 Mar 29;372:n160. doi: 10.1136/bmj.n160.), not the PRISMA statement in 2009. Furthermore, they had better describe they adhered to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 reporting guideline in METHODS.
Reply: Good Comments. We had cited the PRISMA 2020 guideline and make clarification of adherence to the PRISMA guideline.

Reviewer: 3
Dr. Dug Yeo Han, Auckland District Health Board

Comments to the Author:
It is meta-analysis. No suggestion made for statistical analysis.
Reply: Thank you Dr. Dug Yeo Han for your comments.

Reviewer: 4
Mr. AURELIEN HUGUES, Service de médecine physique et réadaptation, hôpital Henry-Gabrielle, Hospices Civils de Lyon

Comments to the Author:
Review
The topic of the research is relevant. The method used is adapted. But the meaning given to the results by the authors is questionable.
Reply: Thank you Mr. AURELIEN HUGUES for your comments. We had made revision accordingly.

General
Avoid “stroke patients” and rather prefer patients with stroke or patients after stroke
Reply: We had revised according to the comments.

Abstract
I2 is missing for short-term effects of home-based rehabilitation combined with UC versus UC alone.
Reply: Thank you for the comments. It had been added.

The first sentence of the conclusion is questionable “Home-based rehabilitation may be an alternative to institution-based rehabilitation and usual care for stroke patients.” Methodological concerns noted later in the conclusion and the request of future researches argue against this statement.
Reply: The first sentence of Conclusion had been deleted.

Introduction
The background is well documented and the justification of research is well established.
Reply: Thank you for the comments.

Method
The search is a bit old (until 31/12/2020). Why? The authors should update the search.
Reply: This study had been updated to December 31, 2021.

The authors do not report which method was applied to limit publication bias? Was the grey literature searched?
Reply: We had revised the Inclusion and Exclusion criteria of Method, and grey literatures were excluded. Publication bias limitation had been added in the last part of Method.

More details could be provided on data extraction, in particular for the point (2).
Reply: We had enriched the information of data extraction.

Please specify the method used to estimate summary effect? Inverse-variance method I think; all variables were continuous outcomes?
Reply: All the data of the studies included in the meta-analyses were continuous data. And the inverse-variance method was used. They had been mentioned in Data analysis of Method.

For the first subgroup analyses, the authors could specify that the level of disability of patients is that in baseline.
Reply: The level of disability could be divided into mild, moderate and severe stroke. It had been mentioned in the fourth paragraph of Data analysis, and Subgroup analysis of Result.

The authors could justify the cut-off chosen (3 points) to consider poor methodological quality of the studies by including a reference.
Reply: Low quality studies could implicate the strength of this review. So, we kept the cut-off as previous studies cited in the Methods-Assessment of methodological quality.

Results
Study characteristics: Maybe the authors could indicate, in brackets, the total number of participants for each category of comparisons. Moreover, the authors could shortly describe the population/participants included in this review (age, time post-stroke, gender, ...)
The table 1 is difficult to read due to its length. It would be better to shorten it a bit. The categorization for the disability level of stroke is not always defined in table 1.
Reply: The study characteristics had been updated and the disability level of stroke had been defined as described in the individual studies. And according to the recommendation of PRISMA 2020 guideline, the summary of details of the home-based intervention in each included study has been extracted according to the Template for Intervention Description and Replication (TIDierR). Table 1 had been revised accordingly.

“Effectiveness of home-based intervention compared with institution-based intervention” part : typo on “a fix-effect” in the last sentence.
Reply: The sentence had been deleted.

Figures 2,3,4 and 5: please adjust the scale of the forest plot.
Reply: The figures had been adjusted.

Discussion and conclusion
Given the results reported, conclude that the effects between home-based intervention and institution-based intervention or usual care might be equivalent seems overestimated. The results such as presented herein do not support these statements.
They seem to be conflicting with the limitations such as reported by the authors. “There is no sufficient

study to determine the effectiveness of home-based intervention comparing with other approaches".
Reply: Discussion and Conclusion had been revised accordingly.

VERSION 2 – REVIEW

REVIEWER	Sheehy, Lisa Mary Bruyère Research Institute
REVIEW RETURNED	31-May-2022

GENERAL COMMENTS	The authors did a good job responding to the reviewers' suggestions. I do not see the revised figures so cannot comment on them. A final review by a native English-speaker would be helpful.

REVIEWER	Banno, Masahiro Seichiryo Hospital, Department of Psychiatry
REVIEW RETURNED	17-May-2022

GENERAL COMMENTS	The authors have revised the manuscript thoroughly. I have no additional comments.